

Introduction of The Power BI Project

Here we have used AdventureWorks2014 sql Database, includes transactional data from 2011 to 2014, basically AdventureWorks2014 cover purchasing to sales, inventory, and manufacturing. For our Power BI demo we mainly focused on sales domain, we created three dashboards, focusing specifically on *Sales*, *Customer Churn*, and *Salesperson Performance*. We loaded tables relevant only to sales to keep the analysis streamlined and aligned with our project objectives.

AdventureWorks categorizes products into four primary groups: *Bikes*, *Components*, *Clothing*, and *Accessories*. The data is structured to allow in-depth analysis by category, showing patterns in sales volume, seasonality, customer behavior, salesperson performance over the four-year span. The company operates across various sales territories, each with distinct revenue and customer trends. The head office is set in Bothell, Washington, United States, with territories spanning North America, Europe, and the Pacific regions (such as Australia).

Our Power BI dashboards provide a clear view of sales trends across territories, helping identify issues and areas for growth. These insights support strategy development to increase sales, reduce customer churn, and improve salesperson performance.

Sales Analysis

Sales By Territory

The dashboard provides an in-depth view of sales trends by territory, year, product category, and continent, alongside a forecasted sales trend. Observing the sharp decline in sales in 2014

In Sales by Territory, territories like Germany, Northeast, and France have the lowest sales.

Our main product category is Bikes. You can see that Bike sales in Germany, Northeast, and France are not very high compared to other regions.

Example: These regions may not be interested in products like Bike due to local preferences or ineffective marketing campaigns, market Saturation

Some products may be more in demand in regions like **Northwest** due to climate or lifestyle preferences.

Sales by year

This bar chart reveals a growth trend from 2011 to 2013, with a peak at 44 million in 2013. However, sales dropped significantly to 20million in 2014, indicating a 54.24% year-over-year decline.

The sharp decline from 2013 to 2014 could result from various factors such as market saturation, demand variations in key product lines, seasonal or increased competition. Especially in categories like Accessories.

New competitors may have entered the market, offering similar or superior products at competitive prices or with innovative features (e.g., electric bikes, customized designs).

If we click on the Bike Category, you can see how sales reflect, how sales drop from 2013 to 2014 it - 52.12%, if we click one by one category, you can realize which category mostly contributes to drop the sales, Mainly component drop sales (-70.27%), Clothing (-56.99%), but accessory drop only (-30.18%) sales from previous year.

Population vs sales

This chart shows the comparison of population and sales rates across different regions. We can see that the Northwest has the highest sales rate, meaning it's doing very well in terms of sales compared to its population size. The Southwest is also performing well, though not as strongly as the Northwest. Canada and Australia are doing moderately well, but other regions like the Northeast, Central, the UK, France, Southeast, and Germany have very low sales rates. This suggests there may be a lot of untapped potential in these areas. AdventureWorks could focus on improving sales in these low-rate regions by creating marketing strategies and products that suit local preferences. It might help to understand local needs better and introduce products that align with those. For example, AdventureWorks could explore offering different products, giving special promotions, or adjusting prices to appeal to these areas. Expanding brand awareness in Europe and North America could also attract more customers. By doing this, AdventureWorks can balance sales across regions and grow its overall market, especially in regions that currently aren't reaching their sales potential.

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Product Category Distribution Trends Over the Years

This area chart shows how different product categories have performed from 2011 to 2014. Accessories were the top-selling category, reaching a peak of 21,000 units in 2013 but dropping slightly to 19,000 units in 2014. Other categories, like Clothing and Components, have stayed fairly steady over the years, with only small changes.

When we look at bicycles, which are the main product for AdventureWorks, we can see a strong interest from consumers. However, there was a noticeable drop in bicycle sales in 2014. This could mean that demand decreased, or there may have been supply chain problems. One reason for the drop could be market saturation. Many people in key areas like North America may already own a bicycle, so fewer new sales are happening.

Since bicycles are built to last, customers don't need to buy them frequently. This limits repeat purchases and affects sales growth. Overall, bicycles have been a big source of income for AdventureWorks, but the

decline in 2014 shows they might need to think of new ways to keep customers interested, such as launching new bicycle models or adding features to attract repeat buyers.

Product Distribution Trends by Year and Territory

- **Insight:** The stacked area chart provides insights into product distribution across territories over the years. While all territories contribute, the Southwest, Canada, and Northwest show a larger share, with a significant decline across territories in 2014.
- **Analysis:** This trend underlines the correlation between territory performance and the overall sales decline. A drop across most regions in 2014 hints at global market factors affecting demand or potential internal distribution issues impacting supply in these territories.
- **Story:** "The simultaneous decline across territories in 2014 suggests that AdventureWorks might be facing a broad market challenge, possibly due to economic conditions or shifts in customer preferences."

Sales by Continent

- **Insight:** The donut chart reveals that North America accounts for 72.42% of total sales, with Europe contributing only 18% and the Pacific region a minor 9.59%.
- **Analysis:** AdventureWorks has a strong market hold in North America but is struggling to capture significant market share in Europe and the Pacific. This imbalance suggests a need to explore new strategies to increase brand presence and sales in these regions.
- **Story:** "The high concentration of sales in North America indicates untapped potential in Europe and the Pacific. Expanding market presence in these regions could provide a boost to overall sales performance." Sales in Europe and Pacific regions are underperforming, especially in categories like Accessories or Clothing, where regional tastes may differ.

Sales Forecast (2014-2024):

- The sales forecast shows a decline after 2014, with a drop from 44M to 20M, indicating a negative trend. The forecast shows uncertainty beyond 2016, suggesting poor growth.
- The dotted line extends beyond 2014 to show the projected sales trend based on historical data.
- The gray shaded area around the forecast represents the confidence interval. This indicates that range within which future sales are expected to fall, with wider intervals in later years reflecting increasing uncertainty.

- If the main product is Bikes, this could imply reduced customer interest or strong competition in this category.
- If you click on Bike, and Component the sales forecast is show like this , that mean in future reduce the customer interest

When you click on the (Select the Categories) from the chart, you can see How Bike forecasting show, we can see bike sales forecasting will be reduce due to Market saturation .

But we select Accessories we can see sales forecasting is good , there will be good demand in future.

Recommendations

Based on the insights from Sales dashboard,

- Since bicycles are the top-selling product but saw a decline in 2014, introducing new models or variants (e.g., electric bikes or improved designs) could rekindle customer interest.
- **Expand Market Presence in Europe and the Pacific:**With a strong foothold in North America, AdventureWorks should explore strategic initiatives, including targeted advertising, partnerships, or region-specific product customization, to increase market share in underperforming regions like Europe and the Pacific.
- **Invest in Market Research for Territory-specific Demand:** To better understand the needs of territories like Germany and the UK, management should conduct detailed market research to identify specific customer preferences and adapt offerings accordingly.
- **Focus on Supply Chain Optimization:** To ensure product availability and reduce potential bottlenecks, evaluate supply chain processes, especially for high-demand items like bicycles, to prevent supply issues in major territories.
- **Implement Dynamic Pricing and Promotions:** To stimulate demand, especially in declining sales periods, consider offering promotions or flexible pricing based on seasonal or regional demand patterns.
- Implement loyalty programs or membership clubs for existing customers to retain them and encourage repeat purchases. This can help maintain a steady sales stream, even during periods of market downturn.
Example: Offer customers rewards or discounts on future Clothing or Accessories purchases for every Bike or Component purchase, creating an ongoing incentive to buy more

Customer Analysis

The dashboard shows significant customer churn analysis in **AdventureWorks** based on **churn rate, customer spending trends, and customer engagement trends** across different regions. Below are eight points highlighting the problems and potential stories behind customer churn, followed by strategies to retain customers for products like **Bikes, Accessories, Clothing, and Components**.

Churn Rate and Churned Customers

The churn rate is currently alarmingly high at 94.53%, which means that a large majority of customers have not made a purchase within a year of their last purchase. In total, 18,735 customers have churned from 2012 to 2014, with this trend increasing significantly each year. The Churned Customers by Year chart reveals a sharp rise in churn from 2012 to 2013, followed by a continued upward trend into 2014. This steady increase indicates that the company is facing a growing challenge in retaining customers over time.

A key factor here is the nature of our main product, bicycles, which are long-lasting and don't require frequent replacement. Once customers purchase a bicycle, they may not need to buy another for several years. This affects repeat sales, as customers primarily return for accessories and components rather than new bicycles. While these add-ons generate some sales, they do not match the revenue of a bicycle purchase, impacting overall retention figures.

Additionally, the market landscape has changed, with new competitors entering and offering similar products. Many of our customers have switched to these competitors, possibly due to attractive pricing, newer models, or unique features like electric bikes or customizable options. This shift has further contributed to our high churn rate, as we are losing customers to other vendors who may be meeting their needs more effectively.

In summary, the high churn rate is due to a combination of product lifecycle factors and increased competition. To address this, the company could consider strategies to encourage repeat purchases, such as introducing loyalty programs, offering upgrade options, or releasing new bicycle models with enhanced features. Additionally, focusing on accessory and component sales, and perhaps offering bundled packages or discounts for returning customers, could help improve retention and reduce churn over time.

Reason for customer leave may be.

Decreasing Customer Satisfaction: Customers may be dissatisfied due to poor service, product quality, or lack of value.

Increased Competition: Competitors may be offering better prices, more features, or improved service, drawing customers away.

Economic Factors: Financial constraints might lead customers to reduce discretionary spending or shift to cheaper alternatives.

Churned Customers and Churn Rate by Territory

The analysis reveals that the Southwest region has the highest customer churn, followed closely by Australia and the Northwest. On the other hand, regions like the Northeast and Central show the lowest churn rates, indicating that customers in these areas are more likely to stay with the company.

This pattern suggests that different regions are facing unique challenges that could be influencing customer retention. For example, areas with higher churn, like the Southwest, may have specific complaints related to certain products. There might be dissatisfaction with the range or quality of Accessories or Clothing, which are popular categories but may not be meeting customer expectations in these regions. Additionally, some of these regions might face strong competition from local brands, which could be pulling customers away.

The regional differences in churn also imply that factors like competition, customer preferences, and economic conditions may vary significantly by location. In some areas, customers may prefer specific products or brands, and in others, there might be more economic pressure or limited disposable income, leading customers to switch to cheaper alternatives.

- **Reasons may by**

- **Regional Competition:** Regions with high churn could face competition, causing customers to switch brands.
- **Service and Product Availability:** The lower churn in certain regions might indicate better service or product availability, whereas higher churn regions may experience stock issues or lack local support.
- **Cultural or Economic Differences:** Customers in some areas may have different expectations or economic pressures impacting their purchasing decisions.

Customer Spend Trend by Year

Adding to this is the trend in customer spending ability. Here, "customer spend" refers to the average purchasing capacity of customers each year. In 2014, the average spend per customer has declined, which could be due to several factors. First, economic conditions in certain regions might have tightened, reducing how much customers are willing to spend on non-essential items like accessories. Additionally, new competitors may have entered the market with lower-priced products, drawing customers away or pressuring the company to offer discounts, ultimately affecting the total spend.

- **Reason may be:**
 - **Customer Disengagement:** As customers become less engaged, they may purchase less frequently or downgrade their purchases.
 - **Limited Product Offerings:** If the product line has not evolved to meet current market trends, customers may be spending less because they don't see value in available products.
 - **Economic Constraints:** Financial downturns or personal economic struggles may lead customers to reduce spending on non-essential items.

Customer Engagement Trends by Territory

1. Engagement Trends in Different Regions

When we look at how customers are interacting with our products in different countries, we see some interesting patterns. In places like **Australia** and **Canada**, customers are very engaged, meaning they often buy and use our products. These countries show some "peaks" or high points in engagement, which suggests that customers there are really interested in what we're offering.

However, in other regions like **France** and **Central**, engagement is more stable, but not as high. Customers in these areas aren't as excited or active with our products, and the numbers stay pretty steady without many spikes.

2. Why Are Customers in Australia and Canada More Engaged?

The high engagement in **Australia** and **Canada** might mean that customers in these regions are very loyal or that our products fit their needs really well. If a customer keeps coming back to buy the same product or a similar product again, it usually means they are satisfied and feel a strong connection to the brand. This kind of loyalty is great because it leads to repeat purchases and word-of-mouth recommendations.

3. Issues with Products in Certain Territories

On the other hand, while customers in **Australia** and **Canada** are engaged, some customers might be **testing** the products but not fully satisfied. We could be seeing problems, especially with products like **Bikes** and **Components**. This means that while customers are trying the products, they might be facing issues like **poor durability** or **low product quality**, which can turn them off and affect their overall satisfaction.

4. Customer Churn in Canada

In **Canada**, we see engagement trends that show people are interested in our products, but **churn** is still a big problem. Churn means customers stop buying from us after some time. Even though they might purchase a few products, they don't stay loyal in the long run. This could be because of issues like **slow delivery**, **poor customer service**, or not enough appealing **after-sales services** (like support after the sale). These customers might find better offers or experiences elsewhere, especially with **Accessories** or **Clothing**.

Recommendations

0. **Enhance Product Line and Innovation:** Introduce new products and update existing ones to meet changing customer preferences. For instance, adding more budget-friendly options or trending products could boost interest and spending.
1. **Personalized Loyalty Programs:**
 - Offer tailored loyalty programs that give points or discounts on future purchases based on customer behavior. Focus on **Accessories** and **Clothing**, where customers are more likely to make repeat purchases.
 - **Example:** Implement discounts on **Bikes** and offer free **Components** servicing after purchase.
2. **Improve After-Sales Service:**
 - Enhancing customer support and after-sales service is crucial for retaining high-value customers, especially in **Southwest** and **Australia**, where churn is high.
 - **Example:** Provide a 24/7 customer service hotline for bike maintenance queries and free servicing within the first year of purchase.
3. **Targeted Marketing for Low-Engagement Territories:**

- Increase brand presence in **France, Germany, and Central** by launching localized campaigns and offering specific products (e.g., winter **Clothing** in Europe).
- **Example:** Partner with local influencers or sponsor biking events in France to boost visibility.

4. Incentivize Returning Customers:

- Offer incentives such as free accessories or discounted components on the next purchase to customers who have previously churned.
- **Example:** For every **Bikes** or **Accessories** purchase, provide a 10% discount on a future **Clothing** purchase.

5. Introduce Subscription Models:

- Implement subscription services for **Accessories** and **Components**, where customers can regularly receive bike parts, new clothing, or maintenance services.
- **Example:** A subscription-based delivery of new bike accessories every quarter with free bike check-ups.

6. Improve Product Variety and Quality:

- Address any customer complaints regarding product quality or variety, especially in **Bikes** and **Components**, to prevent churn in key regions like **Australia** and **Southwest**.
- **Example:** Introduce customizable bikes and components to cater to enthusiasts looking for personalized experiences.

7. Focus on Digital Engagement:

- Engage customers through digital platforms (emails, social media) to keep them informed about new product launches, offers, and loyalty rewards. Focus especially on **Canada** and **Australia**, where engagement is good, but churn is high.
- **Example:** Provide early access to new **Accessories** for loyal customers via email campaigns.

8. Enhance Customer Experience:

- Create a seamless and fast purchasing experience online with fast delivery and clear return policies, especially for regions like **Canada** and **Australia**.
- **Example:** Introduce a same-day or next-day delivery service for popular items like **Components** and **Clothing** to improve customer satisfaction.

Analysis of Salesperson Performance (Problems Identified):

1. Salesperson Contribution by Salesperson:

- The **Sales Contribution by Salesperson** chart shows that a small group of salespersons (Linda, Jillian, and Michael) contribute significantly to overall sales, with each contributing more than 8%.
- However, there is a wide gap between top performers and others like **Stephen, Lynn, and Tete**, who contribute less than 3%, indicating an imbalance in sales distribution across the team.

2. Sales Target Achievement:

- A majority of salespeople are underperforming against their sales quotas. For example, **Jae Pak and Linda Michael** achieved only **their Target**
- Some salespeople, such as **Jose** and **Michael** are performing relatively better, but no one has reached their full sales target.

3. Salesperson Contribution by Territory:

- There is a significant variation in territory performance. For example, the **South west** leads with **22.03%** of total sales, while **Northeast** and **France** are lagging behind at **6.35%** and **6.61%**, respectively.
- Poor performance in these territories could point to regional challenges or the need for better sales strategies.

4. Yearly Sales Performance:

- Sales appear to fluctuate significantly year by year. **2013** shows the highest contribution, but sales sharply decline in **2014**, suggesting external factors or internal challenges that have impacted sales.
- Understanding what led to strong performance in 2013 can help replicate that success in future years.

5. Salesperson Contribution by Year:

- Some salespeople may be coasting on past performance. The consistent top contributors (Linda, Jillian, and Michael) need motivation and performance incentives to maintain or even improve their contributions year on year.
- Identifying why some contributors' performance drops in certain years (e.g., 2014) would be crucial.

6. **Order Processing Performance:** (How frequency order process)

- In terms of order processing, **Jillian Carson** processed the highest number of orders, but her sales achievement percentage is only **47.11%**. This suggests an issue with order Quantity or customer retention.
- Salespeople such as **Syed and Amy** are processing far fewer orders (under 50), raising concerns about workload distribution or motivation.

7. **Customer Churn and Strategy:**

- Since many salespeople are underachieving, this suggests potential customer churn or missed opportunities. Salespeople who are not hitting their quotas may not be retaining customers effectively or upselling.
- Strategies need to focus on **customer retention** and **personalized engagement** to reduce churn.

8. **Regional and Salesperson Imbalance:**

- The imbalance in sales contributions between territories and salespersons suggests that high performers are concentrated in certain regions. Addressing this disparity is critical to improving overall sales performance.

Strategies to Improve Salesperson Performance:

1. **Personalized Sales Training:**

- Provide targeted training for low performers like **Stephen, Lynn, and Tete**. Focus on improving product knowledge, sales techniques, and customer relationship management to help them close more deals.

2. **Territory-Specific Strategies:**

- Conduct a detailed analysis of underperforming regions, such as **Canada and Southwest**, to identify the factors leading to low sales. Adjust marketing efforts or sales strategies to better align with regional customer preferences.

3. **Sales Quota Adjustments:**

- Adjust quotas based on historical performance and territory-specific challenges. For salespeople consistently underachieving their quotas, more realistic targets could boost motivation and performance.

4. **Incentive Programs:**

- Implement an incentive program based on both individual and team achievements. Provide rewards not only for hitting quotas but also for improving customer retention, upselling, and achieving consistent year-on-year growth.

5. **Improved Data-Driven Decision-Making:**

- Utilize the insights provided by the dashboard (e.g., individual performance by year and territory) to make data-driven decisions on resource allocation. Focus on top-performing territories and incentivize salespeople to explore underperforming markets.

6. Focus on Customer Retention:

- Salespeople should prioritize retaining customers and reducing churn. Personalized follow-up programs, customer loyalty incentives, and CRM improvements will help maintain customer satisfaction and long-term relationships.

7. Mentorship and Collaboration:

- Pair top performers, like **Linda and Jillian**, with lower-performing salespeople. This mentorship can provide valuable insights, motivation, and improved teamwork across the salesforce.

8. Product Mix Optimization:

- Analyze which product categories (Bikes, Clothing, Accessories, etc.) are contributing to sales success. Encourage salespeople to focus on high-margin products and cross-sell complementary items to boost overall sales.